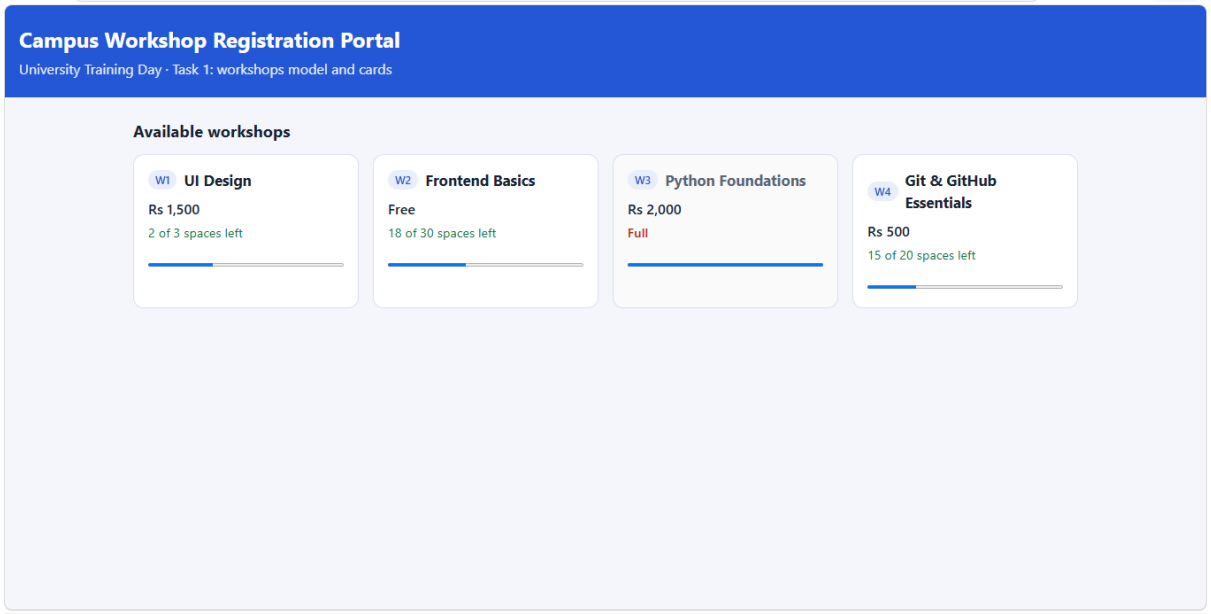
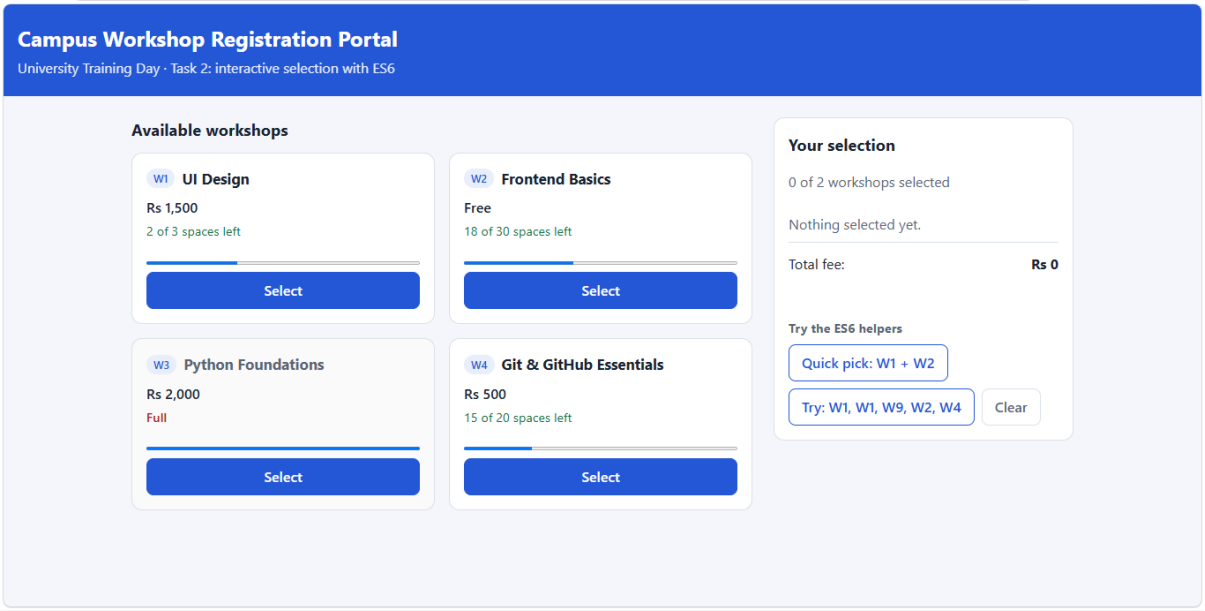
MARYAM JAHANGIR 241949

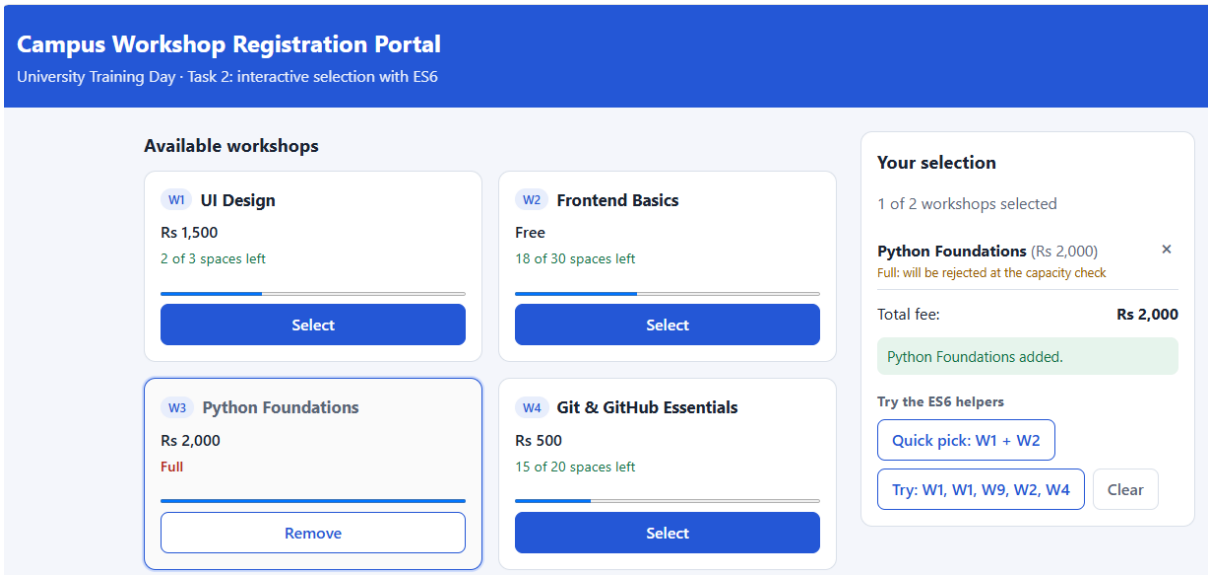
FSWD LAB 6

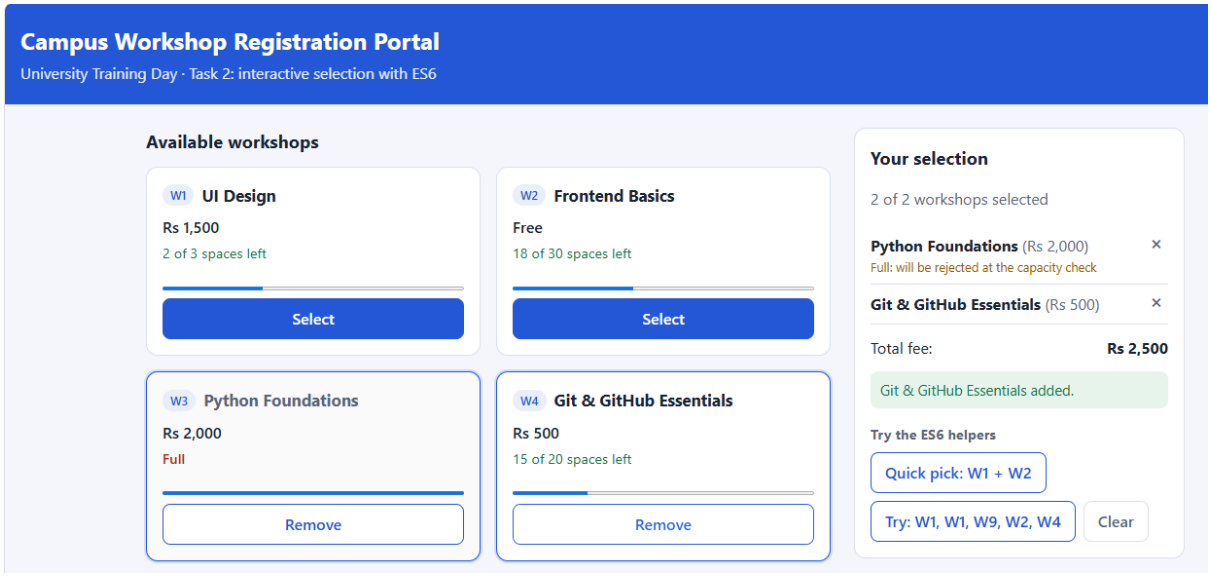
Task 1

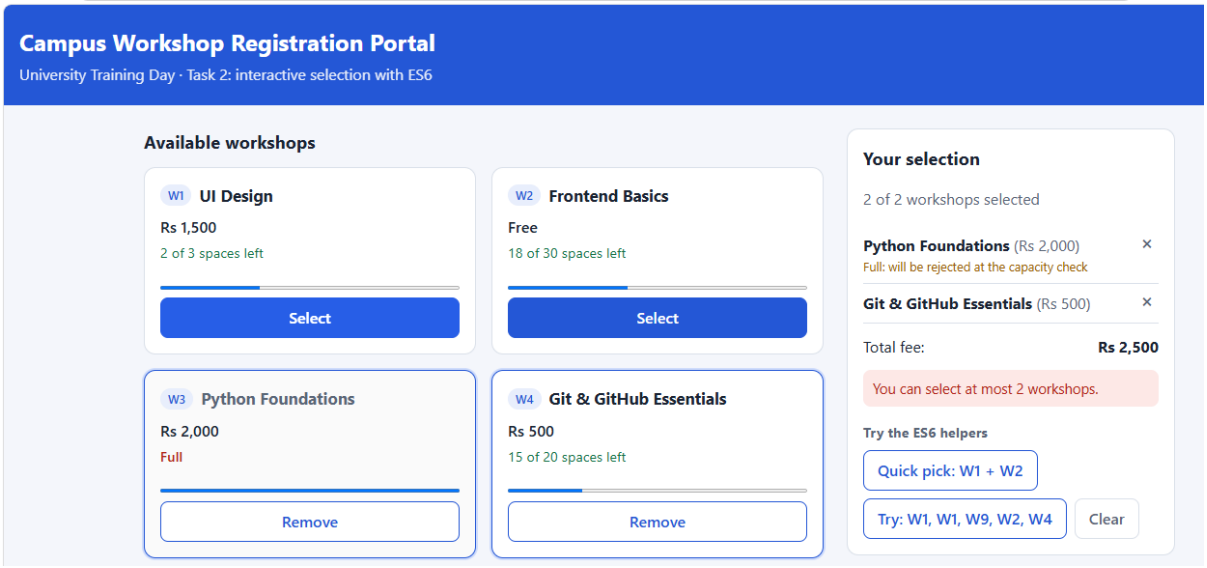

Campus Workshop Registration Portal
University Training Day · Task 1: workshops model and cards
Available workshops
W1
UI Design
Rs 1,500
2 of 3 spaces left
W2
Frontend Basics
Free
18 of 30 spaces left
W3
Python Foundations
Rs 2,000
Full
W4
Git & GitHub Essentials
Rs 500
15 of 20 spaces left

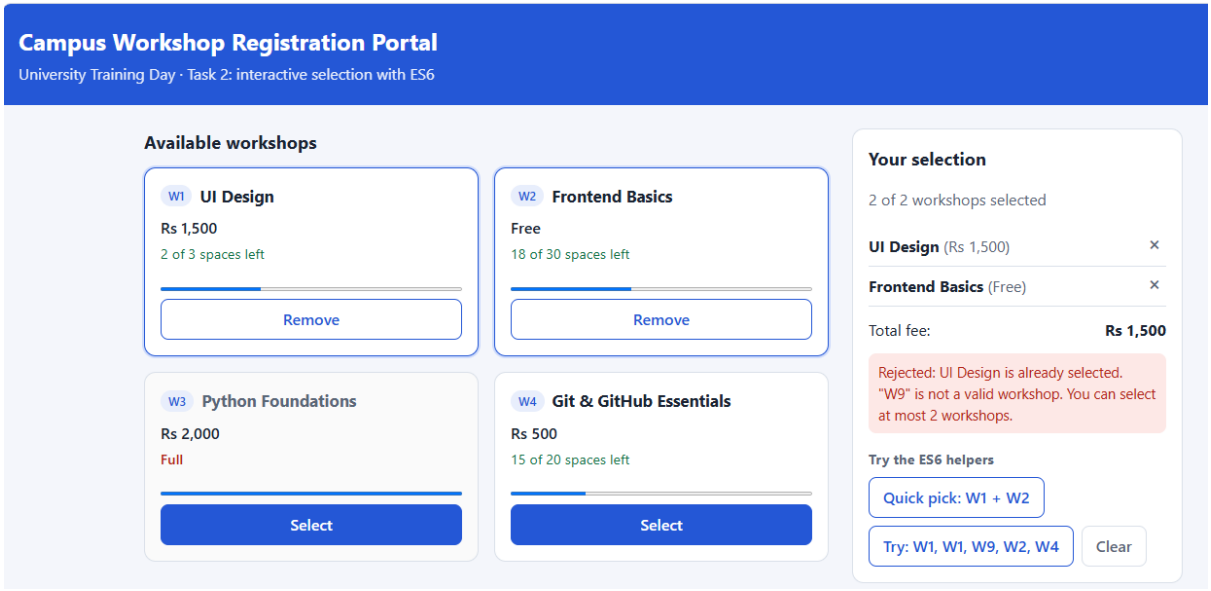
Task 2

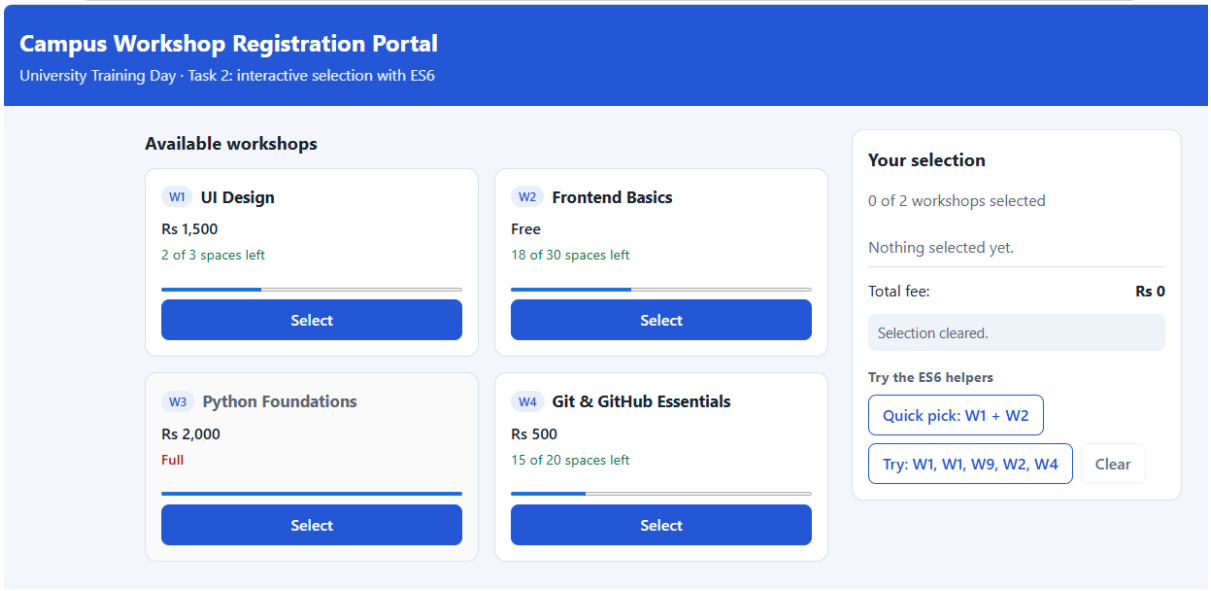

Campus Workshop Registration Portal
University Training Day · Task 2: interactive selection with ES6
Available workshops
W1 UI Design
Rs 1,500
2 of 3 spaces left
Select
W2 Frontend Basics
Free
18 of 30 spaces left
Select
W3 Python Foundations
Rs 2,000
Full
Select
W4 Git & GitHub Essentials
Rs 500
15 of 20 spaces left
Select
Your selection
0 of 2 workshops selected
Nothing selected yet.
Total fee: Rs 0
Try the ES6 helpers
Quick pick: W1 + W2
Try: W1, W1, W9, W2, W4
Clear


Campus Workshop Registration Portal
University Training Day · Task 2: interactive selection with ES6
Available workshops
W1 UI Design
Rs 1,500
2 of 3 spaces left
Select
W2 Frontend Basics
Free
18 of 30 spaces left
Select
W3 Python Foundations
Rs 2,000
Full
Remove
W4 Git & GitHub Essentials
Rs 500
15 of 20 spaces left
Select
Your selection
1 of 2 workshops selected
Python Foundations (Rs 2,000)
Full: will be rejected at the capacity check
Total fee: Rs 2,000
Python Foundations added.
Try the ES6 helpers
Quick pick: W1 + W2
Try: W1, W1, W9, W2, W4
Clear

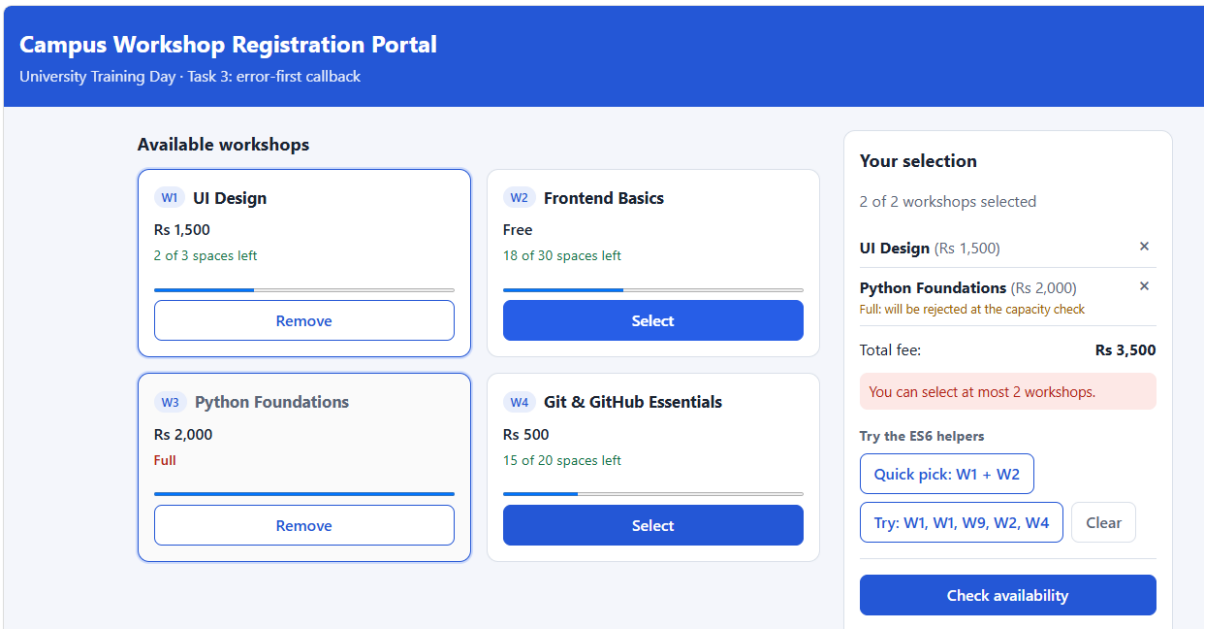

Campus Workshop Registration Portal
University Training Day · Task 2: interactive selection with ES6
Available workshops
W1 UI Design
Rs 1,500
2 of 3 spaces left
Select
W2 Frontend Basics
Free
18 of 30 spaces left
Select
W3 Python Foundations
Rs 2,000
Full
Remove
W4 Git & GitHub Essentials
Rs 500
15 of 20 spaces left
Remove
Your selection
2 of 2 workshops selected
Python Foundations (Rs 2,000)
Full: will be rejected at the capacity check
Git & GitHub Essentials (Rs 500)
Total fee: Rs 2,500
Git & GitHub Essentials added.
Try the ES6 helpers
Quick pick: W1 + W2
Try: W1, W1, W9, W2, W4
Clear


Campus Workshop Registration Portal
University Training Day · Task 2: interactive selection with ES6
Available workshops
W1 UI Design
Rs 1,500
2 of 3 spaces left
Select
W2 Frontend Basics
Free
18 of 30 spaces left
Select
W3 Python Foundations
Rs 2,000
Full
Remove
W4 Git & GitHub Essentials
Rs 500
15 of 20 spaces left
Remove
Your selection
2 of 2 workshops selected
Python Foundations (Rs 2,000)
Full: will be rejected at the capacity check
Git & GitHub Essentials (Rs 500)
Total fee: Rs 2,500
You can select at most 2 workshops.
Try the ES6 helpers
Quick pick: W1 + W2
Try: W1, W1, W9, W2, W4
Clear

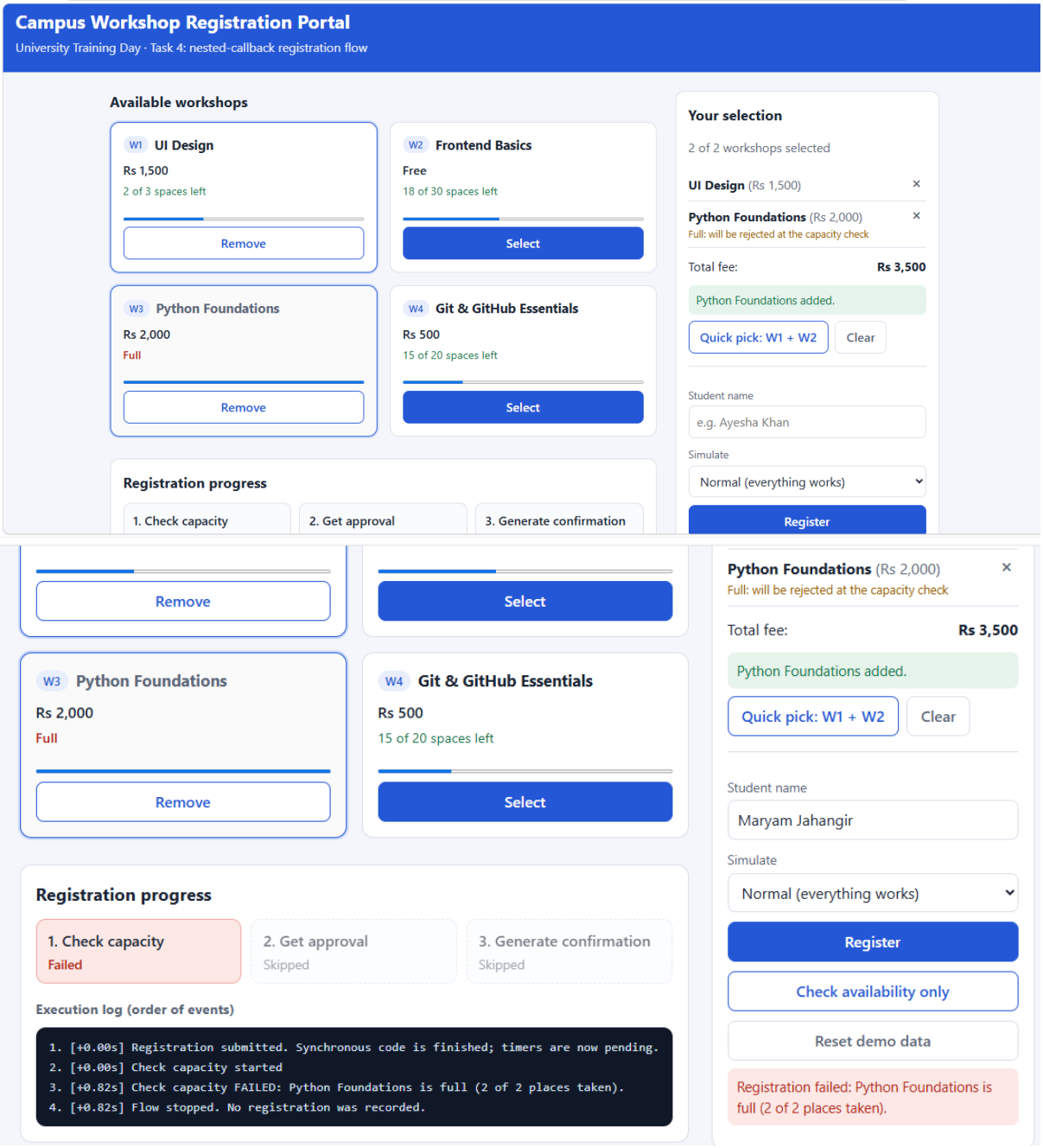

Campus Workshop Registration Portal
University Training Day · Task 2: interactive selection with ES6
Available workshops
W1 UI Design
Rs 1,500
2 of 3 spaces left
Remove
W2 Frontend Basics
Free
18 of 30 spaces left
Remove
W3 Python Foundations
Rs 2,000
Full
Select
W4 Git & GitHub Essentials
Rs 500
15 of 20 spaces left
Select
Your selection
2 of 2 workshops selected
UI Design (Rs 1,500)
Frontend Basics (Free)
Total fee: Rs 1,500
Rejected: UI Design is already selected. "W9" is not a valid workshop. You can select at most 2 workshops.
Try the ES6 helpers
Quick pick: W1 + W2
Try: W1, W1, W9, W2, W4
Clear


Campus Workshop Registration Portal
University Training Day · Task 2: interactive selection with ES6
Available workshops
W1 UI Design
Rs 1,500
2 of 3 spaces left
Select
W2 Frontend Basics
Free
18 of 30 spaces left
Select
W3 Python Foundations
Rs 2,000
Full
Select
W4 Git & GitHub Essentials
Rs 500
15 of 20 spaces left
Select
Your selection
0 of 2 workshops selected
Nothing selected yet.
Total fee: Rs 0
Selection cleared.
Try the ES6 helpers
Quick pick: W1 + W2
Try: W1, W1, W9, W2, W4
Clear

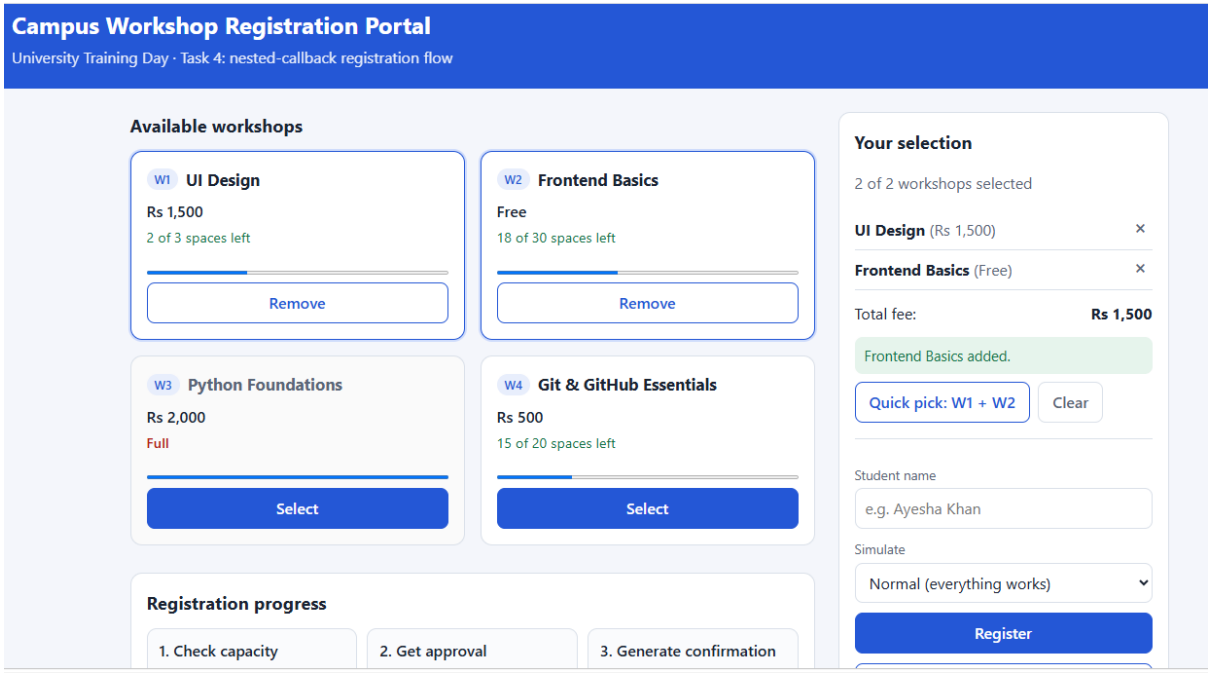
Task 3

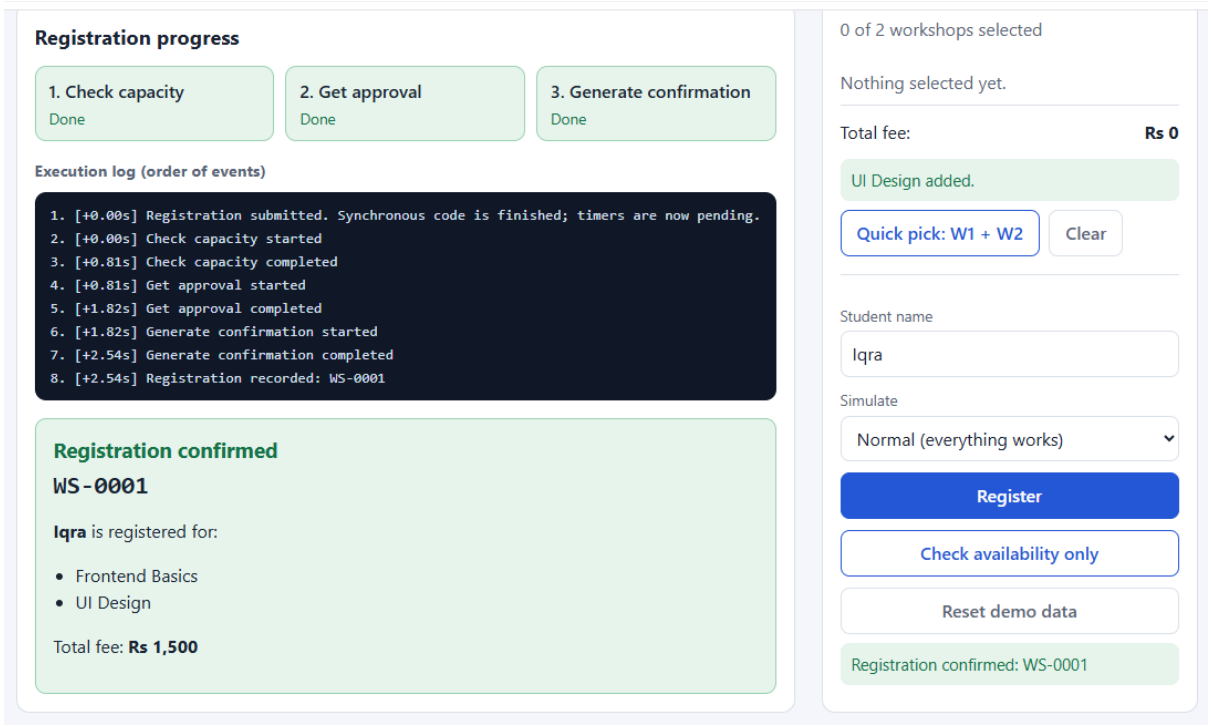

Campus Workshop Registration Portal
University Training Day · Task 3: error-first callback
Available workshops
W1 UI Design
Rs 1,500
2 of 3 spaces left
Remove
W2 Frontend Basics
Free
18 of 30 spaces left
Select
W3 Python Foundations
Rs 2,000
Full
Remove
W4 Git & GitHub Essentials
Rs 500
15 of 20 spaces left
Select
Your selection
2 of 2 workshops selected
UI Design (Rs 1,500)
Python Foundations (Rs 2,000)
Full: will be rejected at the capacity check
Total fee:
Rs 3,500
You can select at most 2 workshops.
Try the ES6 helpers
Quick pick: W1 + W2
Try: W1, W1, W9, W2, W4
Clear
Check availability

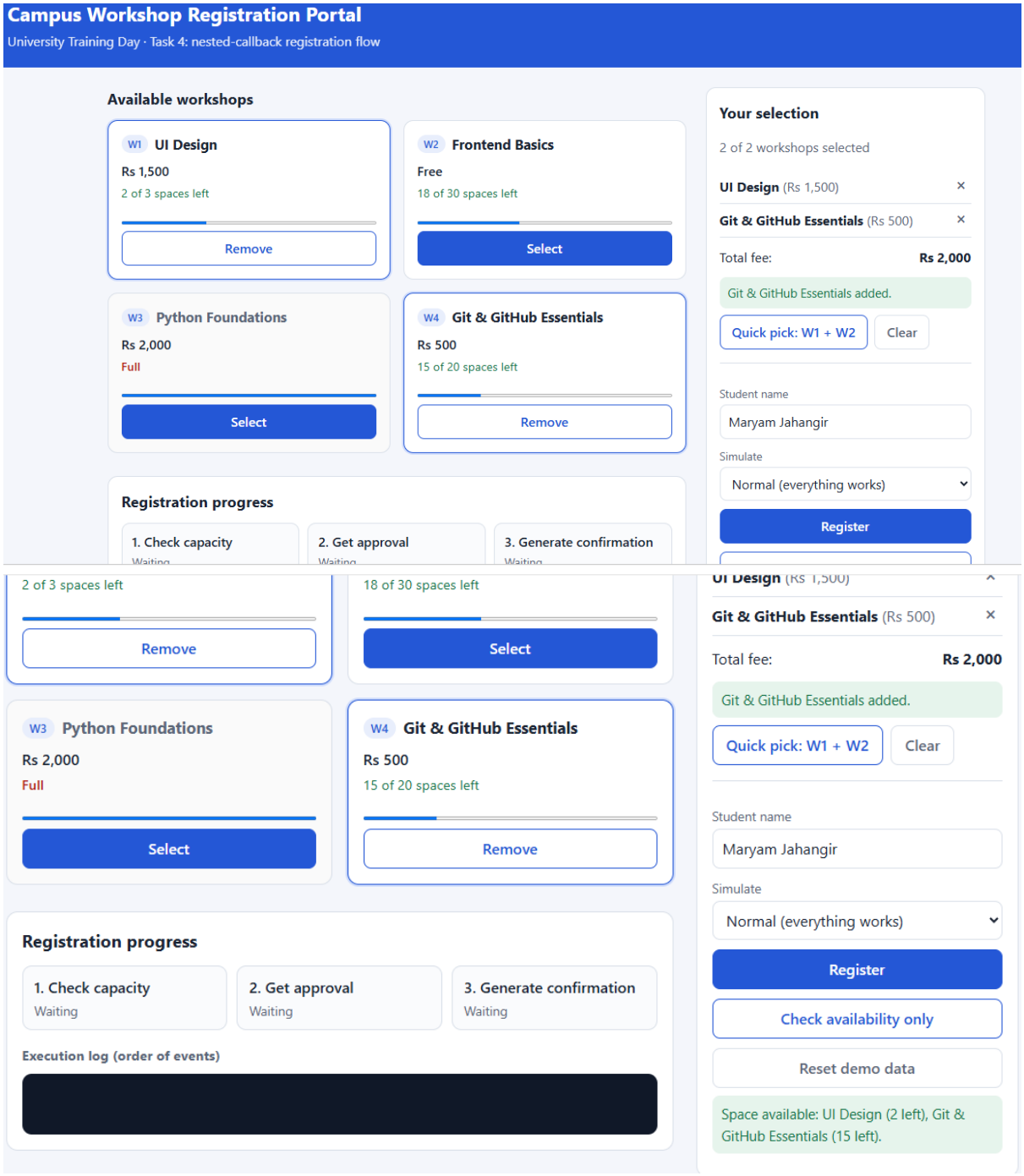
Task 4

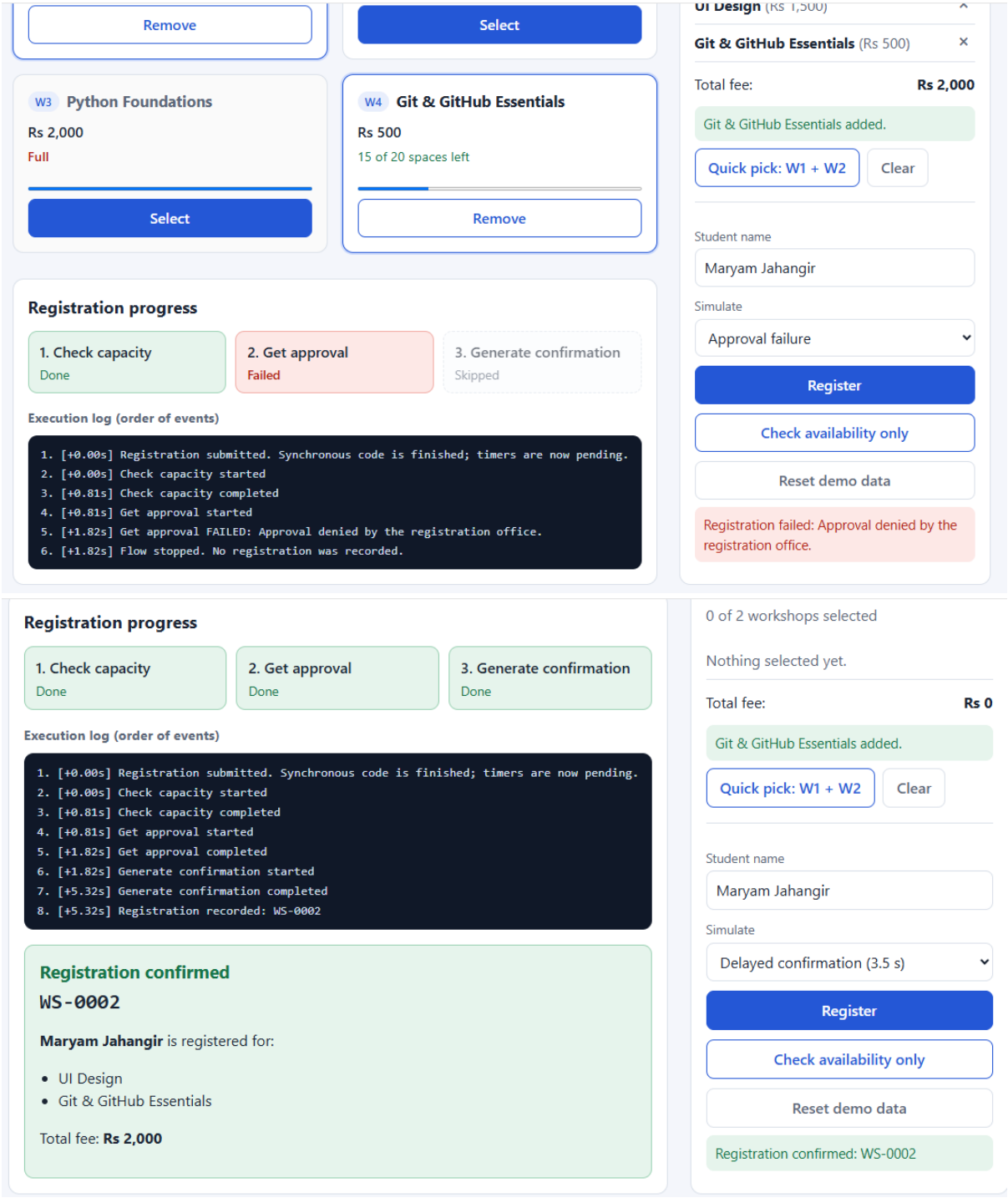

Campus Workshop Registration Portal
University Training Day · Task 4: nested-callback registration flow
Available workshops
W1 UI Design
Rs 1,500
2 of 3 spaces left
Remove
W2 Frontend Basics
Free
18 of 30 spaces left
Select
W3 Python Foundations
Rs 2,000
Full
Remove
W4 Git & GitHub Essentials
Rs 500
15 of 20 spaces left
Select
Registration progress
1. Check capacity
2. Get approval
3. Generate confirmation
Your selection
2 of 2 workshops selected
UI Design (Rs 1,500)
Python Foundations (Rs 2,000)
Full: will be rejected at the capacity check
Total fee:
Rs 3,500
Python Foundations added.
Quick pick: W1 + W2
Clear
Student name
e.g. Ayesha Khan
Simulate
Normal (everything works)
Register

Remove
Select
W3 Python Foundations
Rs 2,000
Full
Remove
W4 Git & GitHub Essentials
Rs 500
15 of 20 spaces left
Select
Registration progress
1. Check capacity
Failed
2. Get approval
Skipped
3. Generate confirmation
Skipped
Execution log (order of events)
1. [+0.00s] Registration submitted. Synchronous code is finished; timers are now pending.
2. [+0.00s] Check capacity started
3. [+0.82s] Check capacity FAILED: Python Foundations is full (2 of 2 places taken).
4. [+0.82s] Flow stopped. No registration was recorded.
Python Foundations (Rs 2,000)
Full: will be rejected at the capacity check
Total fee:
Rs 3,500
Python Foundations added.
Quick pick: W1 + W2
Clear
Student name
Maryam Jahangir
Simulate
Normal (everything works)
Register
Check availability only
Reset demo data
Registration failed: Python Foundations is full (2 of 2 places taken).

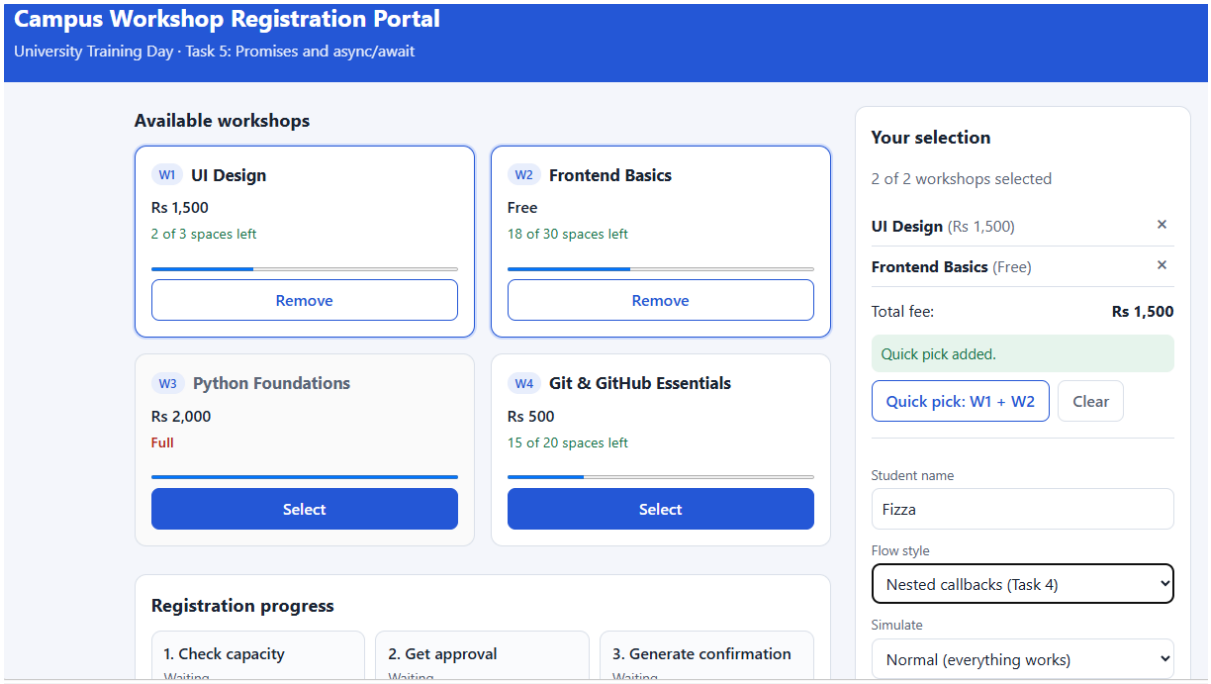

Campus Workshop Registration Portal
University Training Day · Task 4: nested-callback registration flow
Available workshops
W1 UI Design
Rs 1,500
2 of 3 spaces left
Remove
W2 Frontend Basics
Free
18 of 30 spaces left
Remove
W3 Python Foundations
Rs 2,000
Full
Select
W4 Git & GitHub Essentials
Rs 500
15 of 20 spaces left
Select
Registration progress
1. Check capacity
2. Get approval
3. Generate confirmation
Your selection
2 of 2 workshops selected
UI Design (Rs 1,500)
Frontend Basics (Free)
Total fee: Rs 1,500
Frontend Basics added.
Quick pick: W1 + W2
Clear
Student name
e.g. Ayesha Khan
Simulate
Normal (everything works)
Register

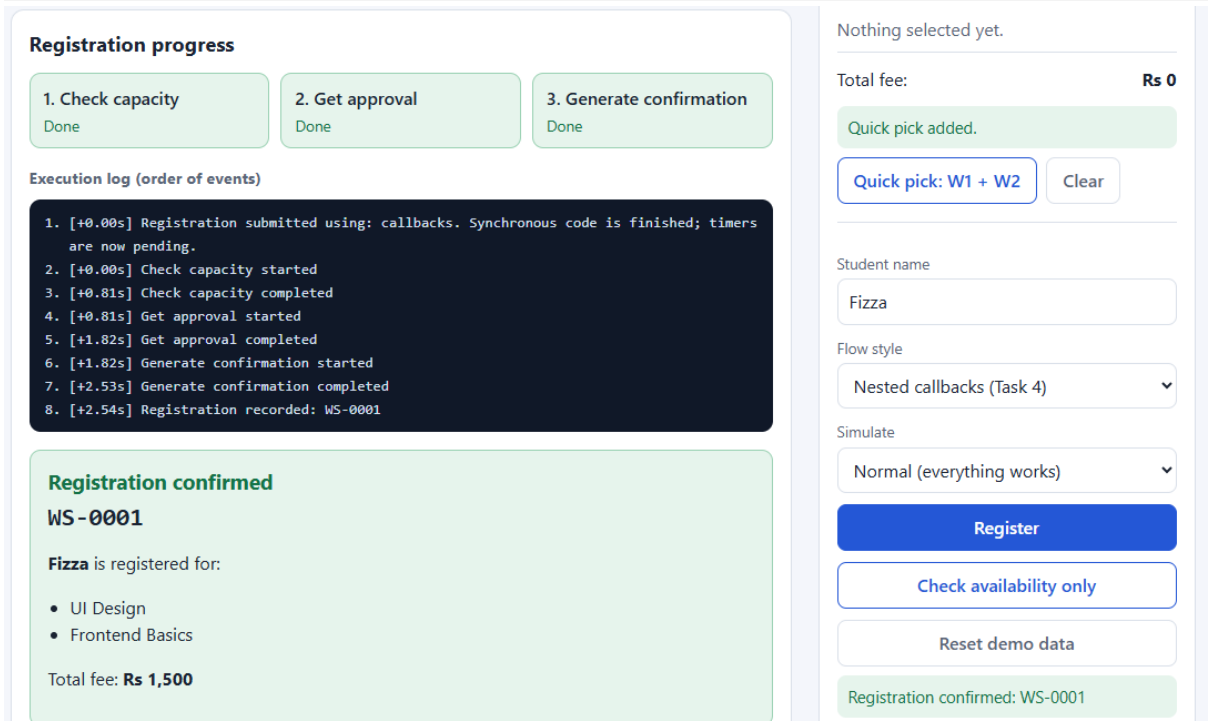

Registration progress
1. Check capacity
Done
2. Get approval
Done
3. Generate confirmation
Done
Execution log (order of events)
1. [+0.00s] Registration submitted. Synchronous code is finished; timers are now pending.
2. [+0.00s] Check capacity started
3. [+0.81s] Check capacity completed
4. [+0.81s] Get approval started
5. [+1.82s] Get approval completed
6. [+1.82s] Generate confirmation started
7. [+2.54s] Generate confirmation completed
8. [+2.54s] Registration recorded: WS-0001
Registration confirmed
WS-0001
Iqra is registered for:
Frontend Basics
UI Design
Total fee: Rs 1,500
0 of 2 workshops selected
Nothing selected yet.
Total fee: Rs 0
UI Design added.
Quick pick: W1 + W2
Clear
Student name
Iqra
Simulate
Normal (everything works)
Register
Check availability only
Reset demo data
Registration confirmed: WS-0001


Campus Workshop Registration Portal
University Training Day · Task 4: nested-callback registration flow
Available workshops
W1 UI Design
Rs 1,500
2 of 3 spaces left
Remove
W2 Frontend Basics
Free
18 of 30 spaces left
Select
W3 Python Foundations
Rs 2,000
Full
Select
W4 Git & GitHub Essentials
Rs 500
15 of 20 spaces left
Remove
Registration progress
1. Check capacity
2. Get approval
3. Generate confirmation
Your selection
2 of 2 workshops selected
UI Design (Rs 1,500)
Git & GitHub Essentials (Rs 500)
Total fee: Rs 2,000
Git & GitHub Essentials added.
Quick pick: W1 + W2
Clear
Student name
Maryam Jahangir
Simulate
Normal (everything works)
Register
Check availability only
Reset demo data
Space available: UI Design (2 left), Git & GitHub Essentials (15 left).
Execution log (order of events)


Remove
Select
W3 Python Foundations
Rs 2,000
Full
Select
W4 Git & GitHub Essentials
Rs 500
15 of 20 spaces left
Remove
Registration progress
1. Check capacity
Done
2. Get approval
Failed
3. Generate confirmation
Skipped
Execution log (order of events)
1. [+0.00s] Registration submitted. Synchronous code is finished; timers are now pending.
2. [+0.00s] Check capacity started
3. [+0.81s] Check capacity completed
4. [+0.81s] Get approval started
5. [+1.82s] Get approval FAILED: Approval denied by the registration office.
6. [+1.82s] Flow stopped. No registration was recorded.
UI Design (Rs 1,500)
Git & GitHub Essentials (Rs 500)
Total fee: Rs 2,000
Git & GitHub Essentials added.
Quick pick: W1 + W2
Clear
Student name
Maryam Jahangir
Simulate
Approval failure
Register
Check availability only
Reset demo data
Registration failed: Approval denied by the registration office.
Registration progress
1. Check capacity
Done
2. Get approval
Done
3. Generate confirmation
Done
Execution log (order of events)
1. [+0.00s] Registration submitted. Synchronous code is finished; timers are now pending.
2. [+0.00s] Check capacity started
3. [+0.81s] Check capacity completed
4. [+0.81s] Get approval started
5. [+1.82s] Get approval completed
6. [+1.82s] Generate confirmation started
7. [+5.32s] Generate confirmation completed
8. [+5.32s] Registration recorded: WS-0002
Registration confirmed
WS-0002
Maryam Jahangir is registered for:
UI Design
Git & GitHub Essentials
Total fee: Rs 2,000
0 of 2 workshops selected
Nothing selected yet.
Total fee: Rs 0
Git & GitHub Essentials added.
Quick pick: W1 + W2
Clear
Student name
Maryam Jahangir
Simulate
Delayed confirmation (3.5 s)
Register
Check availability only
Reset demo data
Registration confirmed: WS-0002

Task 5


Campus Workshop Registration Portal
University Training Day · Task 5: Promises and async/await
Available workshops
W1 UI Design
Rs 1,500
2 of 3 spaces left
Remove
W2 Frontend Basics
Free
18 of 30 spaces left
Remove
W3 Python Foundations
Rs 2,000
Full
Select
W4 Git & GitHub Essentials
Rs 500
15 of 20 spaces left
Select
Registration progress
1. Check capacity
2. Get approval
3. Generate confirmation
Your selection
2 of 2 workshops selected
UI Design (Rs 1,500)
Frontend Basics (Free)
Total fee: Rs 1,500
Quick pick added.
Quick pick: W1 + W2
Clear
Student name
Fizza
Flow style
Nested callbacks (Task 4)
Simulate
Normal (everything works)


Registration progress
1. Check capacity
Done
2. Get approval
Done
3. Generate confirmation
Done
Execution log (order of events)
1. [+0.00s] Registration submitted using: callbacks. Synchronous code is finished; timers are now pending.
2. [+0.00s] Check capacity started
3. [+0.81s] Check capacity completed
4. [+0.81s] Get approval started
5. [+1.82s] Get approval completed
6. [+1.82s] Generate confirmation started
7. [+2.53s] Generate confirmation completed
8. [+2.54s] Registration recorded: WS-0001
Registration confirmed
WS-0001
Fizza is registered for:
UI Design
Frontend Basics
Total fee: Rs 1,500
Nothing selected yet.
Total fee: Rs 0
Quick pick added.
Quick pick: W1 + W2
Clear
Student name
Fizza
Flow style
Nested callbacks (Task 4)
Simulate
Normal (everything works)
Register
Check availability only
Reset demo data
Registration confirmed: WS-0001

Task 6

Demonstrations

Each button resets the demo data, fills in the inputs and runs a registration with the *Flow style* you chose on the right.

1. Successful registration | 2. Full workshop | 3. Approval failure | 4. Delayed confirmation

Registration progress

1. Check capacity	2. Get approval	3. Generate confirmation
Done	Done	Done

Execution log (order of events)

```
1. [+0.00s] Registration submitted using: promises. Synchronous code is finished; timers are now pending.
2. [+0.00s] Check capacity started
3. [+0.82s] Check capacity completed
4. [+0.82s] Get approval started
5. [+1.83s] Get approval completed
6. [+1.83s] Generate confirmation started
7. [+5.34s] Generate confirmation completed
8. [+5.34s] Registration recorded: WS-0001
```

Registration confirmed

WS-0001

Maryam is registered for:

- Frontend Basics

Total fee: **Free**

Acceptance tests

Run acceptance tests

Runs every scenario in all three styles at 50x speed and compares the final state. Takes about a second.

20 of 20 checks passed

Scenario	Style	Result	Detail
Available workshop	callbacks	PASS	ok=true, ref=WS-0001, spaces 2 -> 1
Full workshop	callbacks	PASS	ok=false, "Python Foundations is full (2 of 2 places taken).", stages: start:capacity > fail:capacity
Approval failure	callbacks	PASS	ok=false, "Approval denied by the registration office.", stages: start:capacity > done:capacity > start:approval > fail:approval
Delayed confirmation	callbacks	PASS	ok=true, took 142 ms (slow stage alone is 70 ms), heartbeat ticks=9

Your selection

0 of 2 workshops selected

Nothing selected yet.

Total fee: **Rs 0**

Quick pick: W1 + W2 | Clear

Student name: Maryam

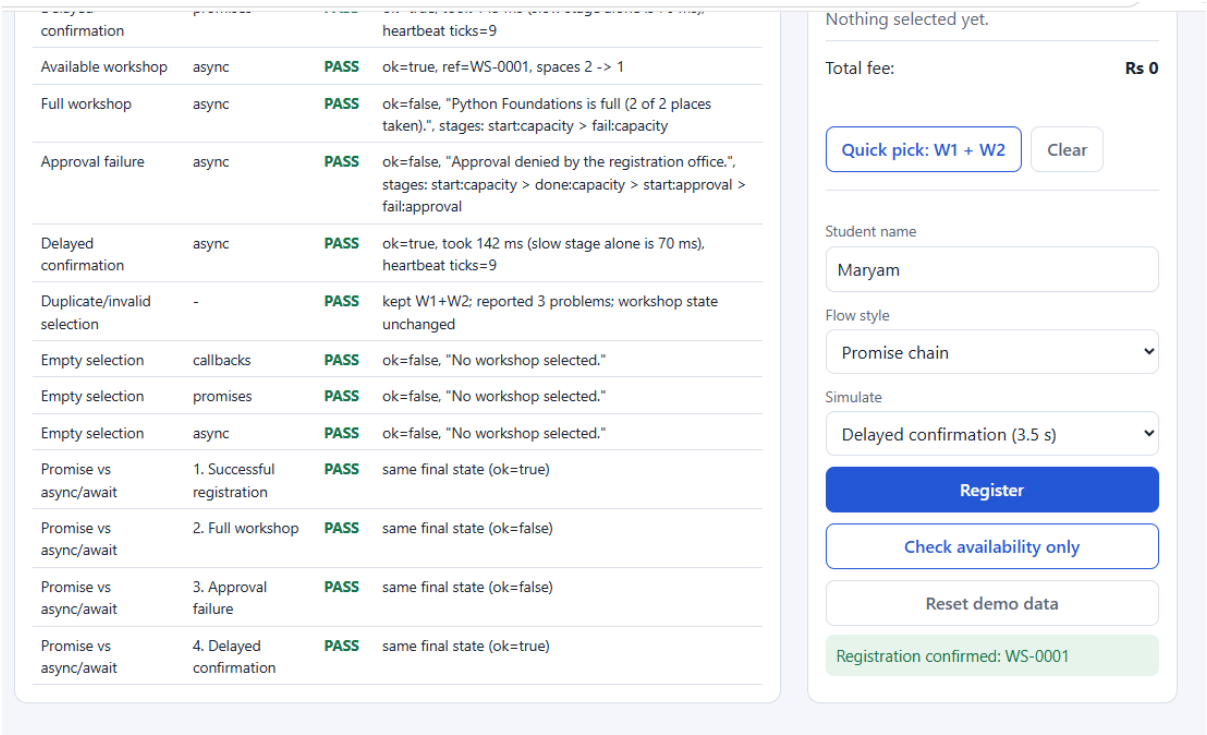
Flow style: Promise chain

Simulate: Delayed confirmation (3.5 s)

Register

Check availability only

Reset demo data



confirmation			heartbeat ticks=9
Available workshop	async	PASS	ok=true, ref=WS-0001, spaces 2 -> 1
Full workshop	async	PASS	ok=false, "Python Foundations is full (2 of 2 places taken).", stages: start:capacity > fail:capacity
Approval failure	async	PASS	ok=false, "Approval denied by the registration office.", stages: start:capacity > done:capacity > start:approval > fail:approval
Delayed confirmation	async	PASS	ok=true, took 142 ms (slow stage alone is 70 ms), heartbeat ticks=9
Duplicate/invalid selection	-	PASS	kept W1+W2; reported 3 problems; workshop state unchanged
Empty selection	callbacks	PASS	ok=false, "No workshop selected."
Empty selection	promises	PASS	ok=false, "No workshop selected."
Empty selection	async	PASS	ok=false, "No workshop selected."
Promise vs async/await	1. Successful registration	PASS	same final state (ok=true)
Promise vs async/await	2. Full workshop	PASS	same final state (ok=false)
Promise vs async/await	3. Approval failure	PASS	same final state (ok=false)
Promise vs async/await	4. Delayed confirmation	PASS	same final state (ok=true)

Concept	Where in the portal	What I learned
Module design (export / import)	workshops.js (data/state), selection.js (rules), callbacks.js / promises.js / asyncawait.js (services and flows), progress.js and ui.js (DOM), app.js (controller)	A module should have one job. Because services never touch the DOM, the same flows run on the page and inside the test suite without changes.
Destructuring	availableSpaces({ capacity, registered }), const { id, name } = workshop, reduce((sum, { fee }) => ...)	It makes the code say which fields a function really needs.
Template literals	Card HTML in ui.js, error messages such as ${workshop.name} is full (...)	Building HTML and messages is far more readable than string concatenation.
Spread	[...selection, id] for a new selection, addMany(selection, ...ids), { ids: [...selection] } for a run snapshot	Creating a new array instead of using push keeps state predictable, and the snapshot stops later clicks from changing a running registration.
Rest parameters	addMany(selection, ...ids)	One function can accept any number of workshop IDs. The same ... means "collect" in a parameter list and "expand" in a call.
Error-first callbacks	checkCapacity, obtainApproval, generateConfirmation	Always check error first and return after reporting it, otherwise the success path also runs.

Concept	Where in the portal	What I learned
Nested callbacks	registerWithCallbacks()	Dependent steps are guaranteed to run in order, but every extra stage adds indentation and another repeated if (error) block.
Promises	promisify(), tracked(), registerWithPromises()	Each .then() must return the next Promise; a rejection skips the remaining .then() calls and lands in one .catch().
async / await	registerWithAsyncAwait() with try / catch / finally	The flow reads top to bottom like synchronous code. await pauses only that function, which the heartbeat counter shows. finally is for logging, never for recording a booking.
Testing failures	tests.js, state snapshots before and after	Checking that failed runs leave the state exactly unchanged is more convincing than checking only the message.
Promises	promisify(), tracked(), registerWithPromises()	Each .then() must return the next Promise; a rejection skips the remaining .then() calls and lands in one .catch().
async / await	registerWithAsyncAwait() with try / catch / finally	The flow reads top to bottom like synchronous code. await pauses only that function, which the heartbeat counter shows. finally is for logging, never for recording a booking.
Testing failures	tests.js, state snapshots before and after	Checking that failed runs leave the state exactly unchanged is more convincing than checking only the message.

2. Comparing the Three Styles

Nested callbacks. A simple building block with no special syntax, but the pyramid of indentation grows with every stage, error handling is repeated at each level, and it is easy to forget a return. Sharing results with other code is awkward.

Promise chains. The chain is flat, errors travel to one .catch(), and Promises can be combined (Promise.all, Promise.race). The cost is discipline: forgetting to return the next Promise silently breaks the ordering, and variables from early steps are hard to reach in later .then() callbacks.

async / await. Easiest to read and debug: normal control flow, try / catch / finally, and early variables stay in scope. It is built on Promises, so I still need to understand them, and independent steps must still use Promise.all or they will run one after another by accident.

3. Which Style I Would Use in a Larger Application

I would use **async/await** for application logic, on top of Promise-returning services, and keep Promise utilities such as Promise.all for steps that can run in parallel. Raw callbacks I would keep only at the edges where a library requires them, and wrap them once with a helper like promisify(), exactly as this project does. In this lab, all three styles produced identical final states for identical inputs (20 of 20 tests passed), so the choice is about maintainability, not behaviour.

4. One Thing I Would Improve

Registrations exist only in browser memory, and two students registering at the same time could both pass the capacity check. A real system would need server-side capacity locking, persistent records and retry/compensation handling.

A. Demonstrations (real-time delays)

#	Style	Selected input	Expected	Actual
1	Callbacks	W1, Normal	3 stages Done, confirmation shown, UI Design spaces 2 -> 1	All 3 stages Done in 2.50 s. Confirmation **WS-0001**. UI Design "1 of 3 spaces left".
2	Promises	W3, Normal	Capacity fails; approval and confirmation never start; no record	Capacity **Failed** at +0.80 s ("Python Foundations is full (2 of 2 places taken)."). Approval and Confirmation **Skipped**. No confirmation card. Counts unchanged.
3	Promises	W1, Approval failure	Capacity Done; approval fails; no confirmation	Capacity Done at +0.80 s, Approval **Failed** at +1.80 s ("Approval denied by the registration office."). Confirmation Skipped. UI Design still "2 of 3 spaces left".
4	async/await	W2, Delayed confirmation	Page stays responsive; result appears only after the delay	Confirmation stage ran from +1.80 s to +5.30 s. While waiting, the heartbeat went 37 -> 41 in 1 s and no confirmation card was visible. After the delay: **WS-0001**, Frontend Basics 18 -> 17 spaces left.

B. Execution Timelines (order of events)

Success, callbacks:

```
1. [+0.00s] Registration submitted using: callbacks. Synchronous code
        is finished; timers are now pending.
2. [+0.00s] Check capacity started
3. [+0.80s] Check capacity completed
4. [+0.80s] Get approval started
5. [+1.80s] Get approval completed
```

```
6. [+1.80s] Generate confirmation started
7. [+2.50s] Generate confirmation completed
8. [+2.50s] Registration recorded: WS-0001
```

Approval failure, promises:

```
1. [+0.00s] Registration submitted using: promises. ...
2. [+0.00s] Check capacity started
3. [+0.80s] Check capacity completed
4. [+0.80s] Get approval started
5. [+1.80s] Get approval FAILED: Approval denied by the registration office.
6. [+1.80s] Flow stopped. No registration was recorded.
```

C. Minimum Acceptance Tests

Scenario	Input	Expected	Actual
Available workshop	W1 (all 3 styles)	Progress completes, confirmation, spaces decrease	PASS x3. WS-0001, spaces 2 -> 1, exactly 1 record
Full workshop	W3 (all 3 styles)	Rejected at capacity; no approval or confirmation	PASS x3. Stages: start:capacity > fail:capacity; state unchanged
Approval failure	W1 + approval failure (all 3 styles)	Capacity OK, approval fails, no confirmation	PASS x3. Stages: start:capacity > done:capacity > start:approval > fail:approval; state unchanged
Delayed confirmation	W2 + slow (all 3 styles)	Responsive; completes only after delay	PASS x3. Elapsed >= slow stage delay; heartbeat ticks 22 (never frozen)
Duplicate / invalid selection	W1, W1, W9, W2, W4	Prevented or reported; consistent state	PASS. Kept W1+W2; reported 3 problems (duplicate, invalid ID, limit); workshop data unchanged
Empty selection	No workshops (all 3 styles)	Rejected; nothing recorded	PASS x3. "No workshop selected."; state unchanged
Promise vs async/await	Each of the 4 demo inputs	Same final state	PASS x4. Same ok, same error text, same snapshot (callbacks also identical)

Total: 20 of 20 checks passed.

Sanity check of the tests themselves: when the approval-failure rule was deliberately broken in a temporary copy, the suite reported 3 FAIL results (one per style), so the tests do detect real faults.